

Mathematics

Stage 8

Paper 1

2025

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

1 Calculate.

$$4\frac{7}{12} - 1\frac{5}{12}$$

Give your answer as a mixed number in its simplest form.

..... [2]

2 Complete these statements.

$$7.2 \times 0.1 = \dots\dots\dots$$

$$7.2 \div 0.1 = \dots\dots\dots$$

[2]

3 Eva draws this table to show equivalent fractions, decimals and percentages. In each of the rows there is **one** incorrect value.

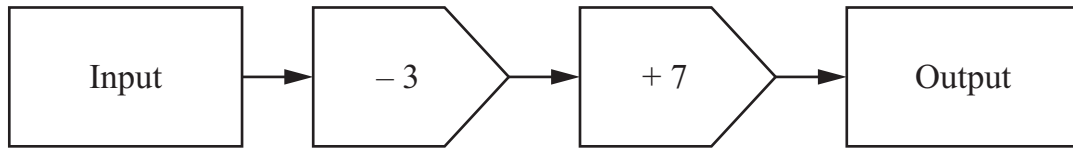
Draw a ring around the incorrect value in **each** row.
One row has been done for you.

Fraction	Decimal	Percentage
$\frac{1}{4}$	2.5	25%
$\frac{3}{10}$	0.03	3%
$\frac{9}{20}$	0.45	4.5%
$\frac{75}{50}$	1.5	15%

[2]

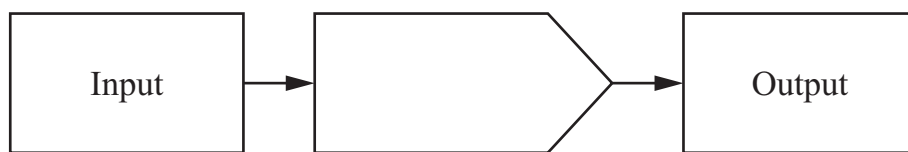
- 4 Mike thinks of a function.

His function can be represented by this function machine involving two operations.



Mike's function can also be represented by a function machine that involves only **one** operation.

Complete this function machine.



[1]

- 5 Calculate.

$$4 \times 1\frac{5}{7}$$

Give your answer as a fraction.

..... [1]

- 6 A regular polygon has exactly 6 lines of symmetry.

Complete these sentences.

The polygon has sides.

The polygon has rotational symmetry order

[2]

- 7 Draw a ring around the fraction that is equivalent to a terminating decimal.

$$\frac{1}{3}$$

$$\frac{1}{5}$$

$$\frac{1}{7}$$

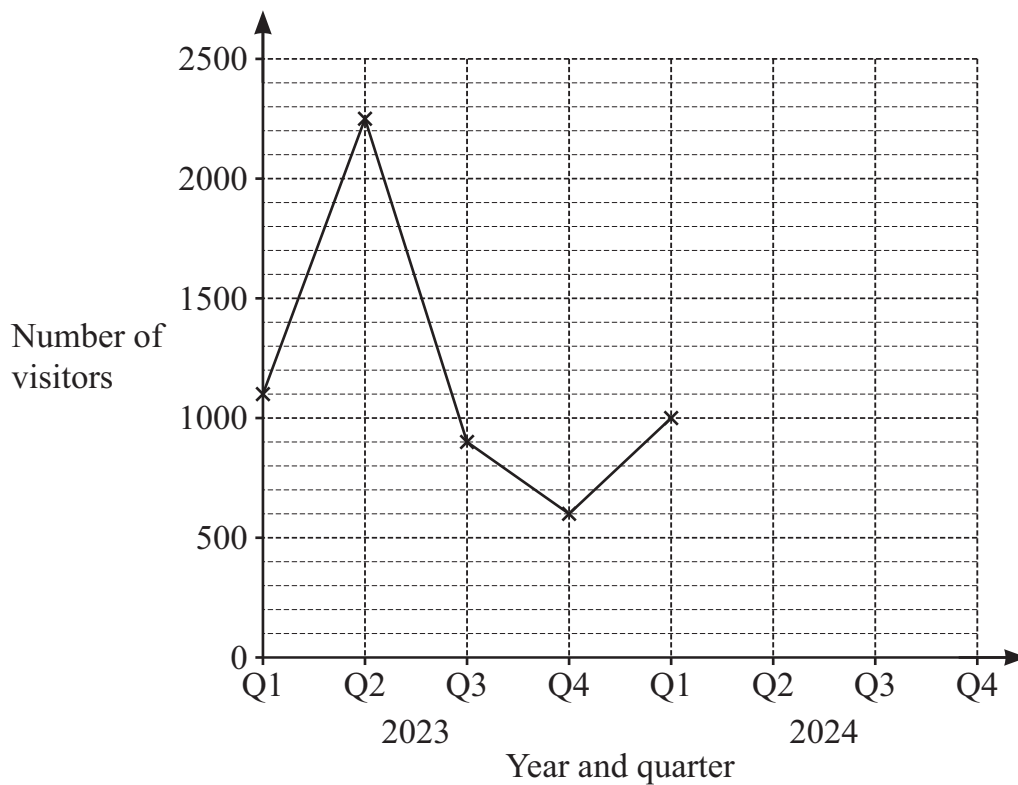
$$\frac{1}{9}$$

[1]

- 8 Lily collects data about the number of visitors to an art gallery.
The table shows some of her data for each quarter (Q) of the years 2023 and 2024

Year	2023				2024			
Quarter	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Number of visitors		2250	900	600	1000	2300	1200	800

- (a) Complete the time series graph to show Lily's data.



[2]

- (b) Use information from the time series graph to complete Lily's table.

[1]

- 9 Carlos has two bags containing coloured counters.

Bag A contains 5 green counters and 10 red counters.

Bag B contains 50 green counters and 100 red counters.

- (a) Carlos takes a counter at random from bag A.

Carlos says,

‘The probability of taking a green counter from bag A is $\frac{1}{3}$ ’

Explain why Carlos is correct.

.....
 [1]

- (b) Carlos says,

‘I am more likely to take a green counter if I take it from bag B rather than bag A.

This is because there are more green counters in bag B.’

Explain why Carlos is **not** correct.

.....
 [1]

- 10 Write down the value of each of these cube roots.

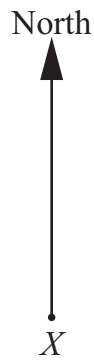
$$\sqrt[3]{-125}$$

.....

$$\sqrt[3]{12^3}$$

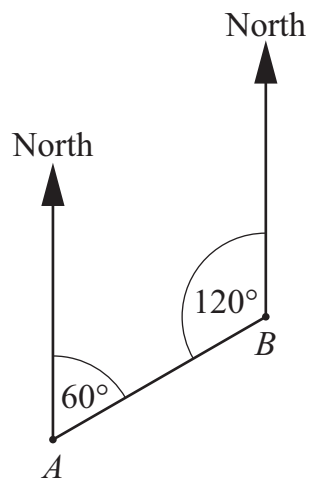
..... [2]

- 11 (a) Draw a bearing of 070° from point X .



[1]

- (b) The diagram shows the position of two ships, A and B .



Find the bearing of A from B .

.....[°] [1]

- 12 (a)** Safia has four spinners, A, B, C and D.
Each of the spinners has four sections numbered 1, 2, 3 and 4

Safia spins each spinner a total of 200 times.

She finds the experimental probability of spinning the number 4 for each spinner.

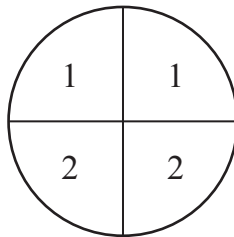
Spinner	A	B	C	D
Experimental probability of spinning a 4	0.22	0.16	0.30	0.27

One of these spinners is **not** fair.

Write down the letter of the spinner this is most likely to be.

..... [1]

- (b)** Oliver has this spinner.



Oliver spins the spinner three times and records the outcomes to make a three-digit number.

Work out how many **different** three-digit numbers he can make.

..... [1]

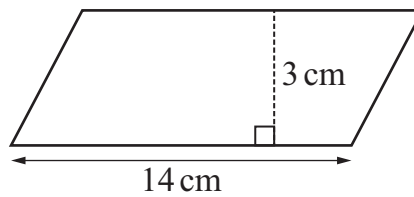
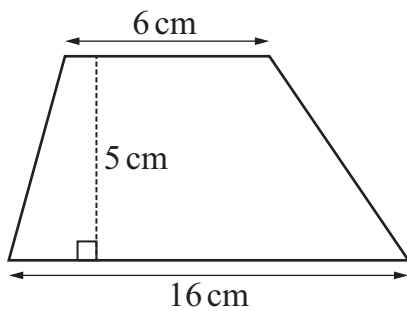
- 13 A sequence has the term-to-term rule subtract $1\frac{1}{2}$

The 3rd term of the sequence is 7

Find the 1st term of the sequence.

..... [1]

- 14 The diagram shows a trapezium and a parallelogram.

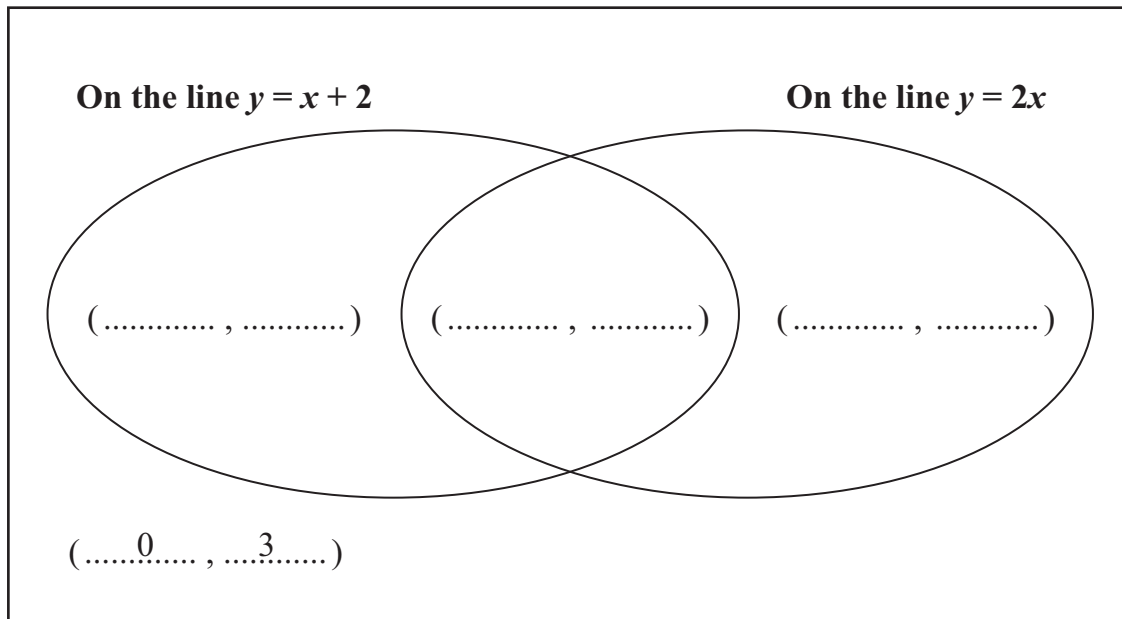


NOT TO
SCALE

Calculate the **difference** between the area of the trapezium and the area of the parallelogram.

..... cm^2 [3]

- 15** Complete the Venn diagram by writing possible coordinates in each of the sections.
One has been done for you.



[3]

- 16** Work out.

$$50 - \sqrt{80 + 8^2} \times 1.5$$

..... [2]

- 17 Jamila draws a pentagon.

The two shortest sides are each of length x cm.

The two longest sides are each three times the length of the shortest sides.

The fifth side is 5 cm longer than the shortest sides.

Find an expression, in terms of x , for the perimeter of the pentagon.

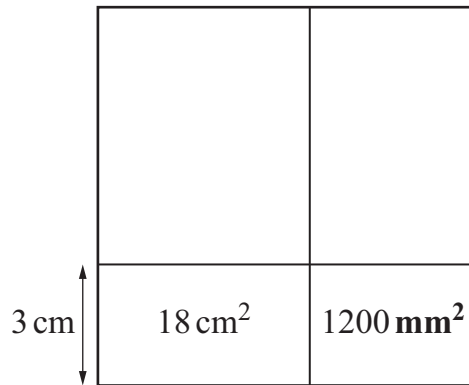
Give your answer in its simplest form.

..... cm [2]

- 18 Samira draws this square.

She draws two straight lines inside the square to make four rectangles.

The areas of the two smallest rectangles are written inside each rectangle.



NOT TO
SCALE

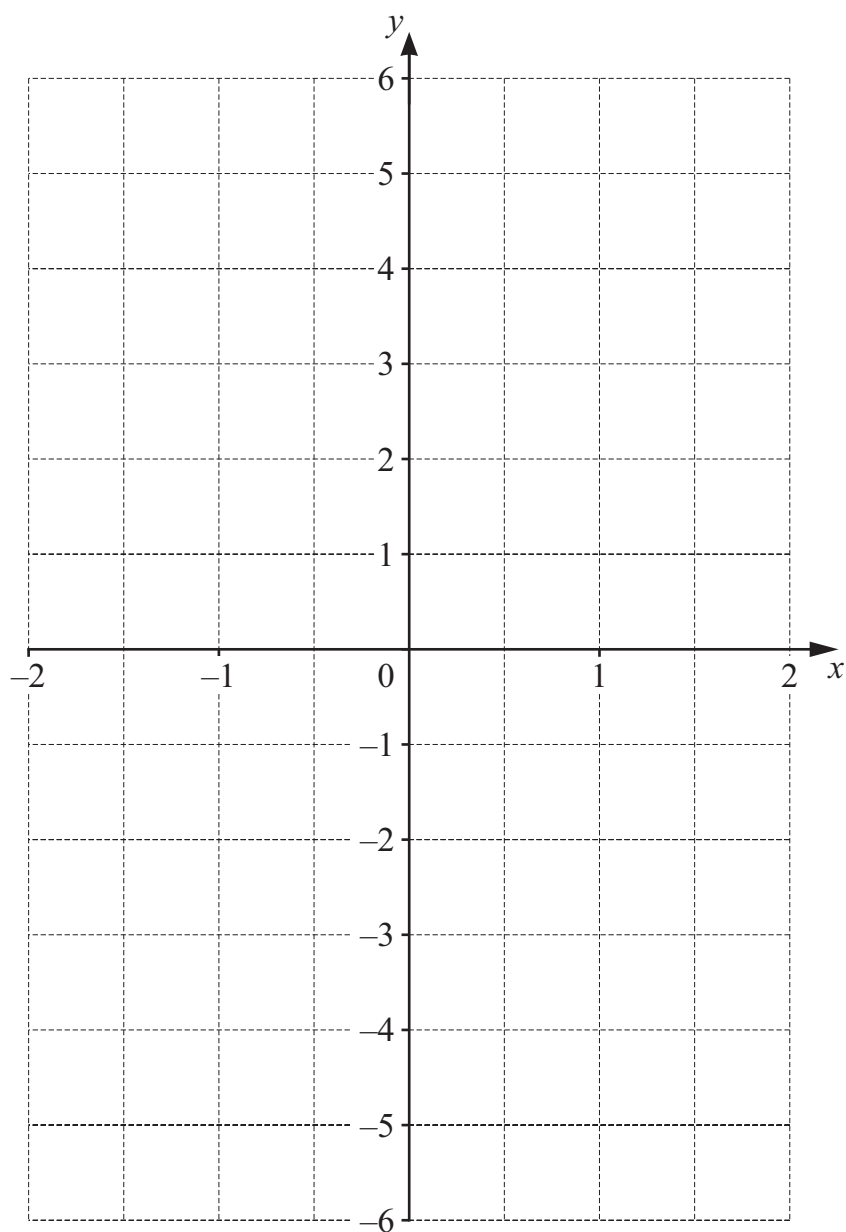
Work out the length of one side of the square.

Give your answer in centimetres.

..... cm [2]

- 19 Draw the graph of $y = 2x + 1$ on the grid.
You may use the table of values to help you.

x	-2	0	2
y			



[3]

20 A point P has coordinates $(-3, 9)$.

- (a)** The point P is translated by the vector $\begin{pmatrix} 8 \\ 2 \end{pmatrix}$.

Find the coordinates of the image of point P .

(..... ,) [1]

- (b)** The point P is reflected in the line $x = 2$

Find the coordinates of the image of point P .

(..... ,) [1]

- (c)** The point P is reflected in the line $y = k$
The coordinates of the image of point P are $(-3, 5)$.

Find the value of k .

$k =$ [1]

21 Write one pair of brackets in this calculation to make it correct.

$$0.6 - 0.2 \times 2 + 0.1 = 0.18$$

[1]

22 Chen has a box containing coloured pens.

There are 9 times as many blue pens in the box as pens that are **not** blue.

Chen picks a pen at random from the box.

Work out the probability the pen Chen picks is **not** blue.

..... [1]

23 Complete these sentences about percentage change and absolute change.

One has been done for you.

To increase something by 60 % you multiply by the decimal 1.6

To increase something by 0.5% you multiply by the decimal

To increase something by % you multiply by 5

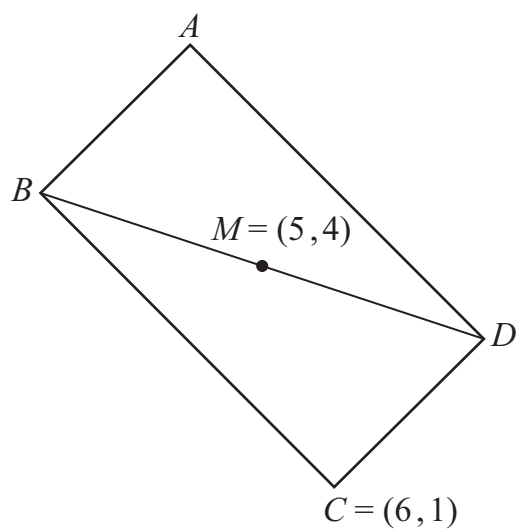
When 400 kg is decreased by 50% the absolute change is kg.

[2]

24 The diagram shows a rectangle $ABCD$.

M is the midpoint of BD .

The coordinates of M and C are shown on the diagram.



NOT TO
SCALE

Find the coordinates of A .

$A = (\dots\dots\dots, \dots\dots\dots)$ [2]

25 Solve.

$$2(5x + 1) = 2 - 5(4x - 3)$$

$$x = \text{.....} [3]$$

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced annually and is available to download at <https://lowersecondary.cambridgeinternational.org/>

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.